CV OF GUILLERMO VILLANUEVA BENITO

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CONTACT

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PERSONAL INFORMATION

· Birth date: May, 12th of 1998.
· Birth place: Zaragoza, Spain.
· Citizenship: Spanish.
· Occuppation: PhD student.

SUMMARY

I am a PhD student in Artificial Intelligence for Healthcare at the Barcelona Supercomputing Center (BSC-CNS) and the Barcelona Institute for Global Health (ISGlobal). My research focuses on developing deep learning and vision-language models for medical imaging analysis, automated diagnosis, and clinical decision support.

I hold a double degree in Mathematics and Engineering Physics from the Interdisciplinary Higher Education Center (CFIS) at the Universitat Politècnica de Catalunya (UPC).

My research interests lie at the intersection of artificial intelligence, medical imaging, and healthcare, with a particular emphasis on translating AI research into real-world clinical applications and impact.

EDUCATION

· **PhD in Artificial Intelligence for Healthcare** *Dec 2023 - Present*
Barcelona Supercomputing Center (BSC-CNS)-Barcelona Institute for Global Health (ISGlobal)
Digital Health Unit. Barcelona, Catalonia, Spain.

– Develop deep learning and vision-language models for medical imaging analysis, automated diagnosis, and clinical decision support.

· **Master's Degree in Advanced Mathematics and Mathematical Engineering. (MAMME)** *Sep 2021 - Sep 2022*
Universitat Politècnica de Catalunya (UPC) - BarcelonaTECH
School of Mathematics and Statistics (FME). Barcelona, Catalonia, Spain.

– Relevant Coursework: Operations Research, Mathematical Models, Machine Learning, Numerical Methods, Statistical Learning.

· **Bachelor's degree in Mathematics.** *Sep 2016 - Jul 2021*
Universitat Politècnica de Catalunya (UPC) - BarcelonaTECH & CFIS
School of Mathematics and Statistics (FME). Barcelona, Catalonia, Spain.

– As part of the CFIS interdisciplinary program to course two degrees simultaneously.

· **Bachelor's degree in Engineering Physics.** *Sep 2016 - Jul 2021*
Universitat Politècnica de Catalunya (UPC) - BarcelonaTECH & CFIS
Barcelona School of Telecommunications Engineering (ETSETB). Barcelona, Catalonia, Spain.

– As part of the CFIS interdisciplinary program to course two degrees simultaneously.

RESEARCH AND WORK EXPERIENCE

· **AI/ML Research Scientist.** *Nov 2022 - Dec 2023*
Barcelona Institute for Global Health (ISGlobal). Barcelona, Catalonia, Spain.

– Analyze biomedical data and develop explainable machine learning models and computer vision techniques for research publications.

· **Operations Research Intern** *Mar 2022 - Aug 2022*
Paack. Barcelona, Catalonia, Spain.

– Data analysis and forecasting. Logistic and supply chain related mathematical optimization models. Collect business requirements and deliver easy to understand data analysis, results, and recommendations to stakeholders.

· **Visiting Researcher (Master's Student).** *Feb 2022 - Sep 2022*
New Jersey Institute of Technology (NJIT) - Rutgers University.
Federated Department of Biological Sciences and NJIT institute of Brain and Neuroscience Research (IBNR).

– Research work on computation neuroscience. Analyze behavioral and neural data from the cerebellum and the basal ganglia (major structures of the brain) to produce a first-generation model of dopamine reward-based learning.

· **Visiting Researcher (Bachelor's Student).** *Sep 2020 - Jul 2021*
New Jersey Institute of Technology (NJIT) - Rutgers University.
Federated Department of Biological Sciences and NJIT institute of Brain and Neuroscience Research (IBNR).

– Research work on computation neuroscience. Analyze mathematical models in neuroscience to study the problem of parameter estimation unidentifiability.

PUBLICATIONS

· **Villanueva Benito, G.**, Descalzo, M. L., Pujadas, S., Fernandez, J., Calandrelli, M., and Petrone, P. (2026). *CineScribe: LLM-based detection and mitigation of ambiguity in cine cardiac magnetic resonance reports.* Computers in Biology and Medicine.

· Pedreira-Rincón, J., Balerdi-Sarasola, L., **Villanueva, G.**, Fleitas, P. E., Jimenez, A., Mayor, A., et al. (2026). *pLDH to identify severity in imported malaria: Implementing smartphone video analysis for rapid clinical decision-making.* Biosensors and Bioelectronics.

· Mondal, Y., **Villanueva Benito, G.**, Pena, R. F. O., and Rotstein, H. G. (2025). *Postsynaptic frequency filters shaped by the interplay of synaptic short-term plasticity and cellular time scales.* Journal of Computational Neuroscience.

· **Villanueva Benito, G.**, Goldberg, X., Brachowicz, N., Castaño-Vinyals, G., Blay, N., Espinosa, A., et al. (2024). *Machine learning for anxiety and depression profiling and risk assessment in the aftermath of an emergency.* Artificial Intelligence in Medicine.

CONFERENCES AND OTHER SCIENTIFIC EVENTS

· **ESC Digital & AI Summit 2026, Basel, Switzerland.** *Nov 2026*
Poster contribution (presented by M. L. Descalzo).
Automated regional wall motion abnormality detection in cine cardiac MRI using LLM-assisted annotation and video deep learning.

· **MICCAI 2026, Strasbourg, France.** *Sep 2026*
Conference attendance.

· **AIHealth 2026, Valencia, Spain.** *Mar 2026*
Oral presentation (Best Paper Award).
CineScribe: LLM-Based Detection and Mitigation of Ambiguity in Cine Cardiac Magnetic Resonance Reports.

· **ESC Congress 2025, Madrid, Spain.** *Aug 2025*
Oral presentation contribution (presented by M. Calandrelli).
CineScribe: AI-driven standardization of cardiac motion reports in cine MRI.

· **Women in Data Science (WiDS) Barcelona 2023, Barcelona, Spain.** *Oct 2023*
Poster presentation.
Machine learning for anxiety and depression profiling in the aftermath of an emergency.

· **30th Annual Computational Neuroscience Meeting (CNS*2021), Online.** *Jul 2021*
Poster presentation.
Neuronal oscillations level sets for activity constancy: from single neurons to networks.

MENTORSHIP

· **Oishik Ray** *Mar 2026 - Sep 2026*
Master's Thesis, InterMaths Erasmus Mundus.
A Segmentation-based Approach to Automated Regional Wall-Motion Abnormality Assessment from Cine CMR.

· **Ricard Garcia Isern** *Mar 2025 - Sep 2025*
Bachelor's Thesis, Bioinformatics.
CardIA: A Deep Learning-Based Workflow for LVEF Extraction from 4D CMR Imaging. Best Thesis Award.

FELLOWSHIPS AND ACHIEVEMENTS

· **Participation in the 2021 Dana Knox Student Research Showcase.** *Apr 2021*

– As a visiting research at the New Jersey Institute of technology (NJIT), I was nominated and selected by the screening committee to participate in The Dana Knox Student Research Showcase. Participants were given a time slot to present their projects virtually to the assigned judges and guests.

· **CFIS Exchange student fellowship.** *Feb 2020*

– A six-month stay at the Mathematical Biology and Computational Neuroscience group in the Federated Department of Biological Sciences at the New Jersey Institute of Technology (NJIT) and Rutgers University and the NJIT Institute of Brain and Neuroscience Research (IBNR).

· **Tuition and Housing Scholarship** *Sep 2016 - Jul 2020*

– Scholarship covering the price of a Bachelor's degree and housing in Barcelona as part of the double-degree program at CFIS.

· **CFIS membership.** *Sep 2016*

– Acceptance to the CFIS Program.

· **Honors in Baccalaureate.** *Jun 2016*

LANGUAGES AND TECHNICAL SKILLS

· **Languages**

- **Spanish:** Native.
- **English:** C1 (UOC, 2021); IELTS Academic 6.5 (2020); Cambridge FCE B2/C1 (2015).
- **Catalan:** Good oral and written comprehension.

· **Programming & Technical Skills**

- **Programming languages:** Python, PyTorch., C++, MATLAB, R, SQL.